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PAUT Corrosion Mapping - 4-RA-101-06 and 6-RA-103-11

Scope and Responsibilities

This report details the Phased Array Ultrasonic Testing (PAUT) corrosion mapping performed at the Hamilton Plant within the ATU-100. The scope includes high-resolution encoded scanning of the 4-RA-101 elbow at 4-RA-101-06 and the straight section at 6-RA-103-11. The objective is to characterize localized wall loss following initial thickness surveys that indicated thinning in the rich amine service piping.

Marcus Chen served as the lead UT technician responsible for data acquisition and preliminary analysis. David Park provided the Level III oversight and final technical review of the encoded data sets. All examinations were performed under QSCP-NDT-001 Rev 3, ensuring compliance with the updated digital export and calibration requirements for the ATU-100 unit.

Reference Standards and Equipment

The inspection methodology and data evaluation criteria were established in accordance with ASME BPVC Section V. The PAUT system utilized for this survey was an Olympus OmniScan X3 1022157. The unit's performance and linearity were verified against LNDE-CAL-2024-0921, which confirms the annual calibration status of the instrument.

This assessment was triggered by findings in LNDE-RPT-20241025-031-R1, which identified areas of concern during the five-year survey. Furthermore, the specific investigation into 4-RA-101-06 was mandated by QSCP-NCR-2024-031, which documented a nonconformance regarding wall thickness measurements falling below the calculated retirement limits.

Examination Results for 4-RA-101-06

The 4-RA-101-06 CML is located on a 114.3 mm OD A106 Gr.B elbow. The design parameters for this line are 862 kPa(g) and 82 °C. PAUT mapping confirmed a minimum wall thickness of 3.55 mm. This value is below the specified t-min of 3.81 mm. Because the measured thickness is less than the retirement thickness, the remaining life for this component is considered expired.

The localized nature of the thinning at this CML suggests accelerated corrosion within the rich amine circuit. Per site rules, since the reading is below t-min, the nonconformance remains active. The data indicates that the short-term corrosion rate has exceeded the capacity of the nominal 8.56 mm wall to maintain integrity through the current interval.

Examination Results for 6-RA-103-11

The 6-RA-103-11 CML was inspected on the 168.3 mm OD rich amine header. This A106 Gr.B piping has a nominal thickness of 10.97 mm and a t-min of 4.57 mm. The PAUT scan identified a minimum remaining wall of 5.82 mm. Based on the larger of the long-term and short-term corrosion rates, the remaining life is calculated at 2.1 years.

Under the site rules for Class 2 amine service, the next thickness reading must be scheduled at no more than half the remaining life. Consequently, the next inspection for 6-RA-103-11 is required within 1.05 years. This interval is more restrictive than the standard 3-year RBI interval for rich amine lines, due to the observed degradation rate.

Thickness readings, measured 2024-12-10

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-06	8.56	3.81	2024-11-04	3.59	3.55	0.426	0.659	expired	immediate	< t-min; gauge photo attached

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-11	10.97	4.57	2024-11-04	5.92	5.82	0.419	0.600	2.1	2025-12-25	RL/2 < interval; gauge photo attached

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2024-12-10	_____
Reviewed	David Park	Level III / Quality Manager, SNT-TC-1A UT Level III	2024-12-10	_____